Amit Amit

Data Scientist | Computer Vision and Applied AI

Gotha, Germany | linkedin.com/in/amit-tech
Portfolio and case studies | GitHub: amit-tech-datascience | Earlier work: ak06645

Data Scientist at Sedeco Imaging working on camera-based detection, robotic handling and efficient edge deployment. M.Sc. in Data Science from TU Braunschweig, with research experience in scientific machine learning, biomedical data analysis and aerospace applications. Earlier experience in Linux infrastructure, automation and SQL reporting.

Technical skills

Modeling: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, anomaly detection, segmentation, physics-informed neural networks.
Data and engineering: Python, SQL, NumPy, Pandas, Git, Docker, Linux, Django, TypeScript/Next.js.
Vision and edge: OpenMV, Raspberry Pi, Luckfox, Hikrobot camera integration.

Professional experience

Data Scientist | Sedeco Imaging

Gotha, Germany | October 2025 - Present

- Lead the technical design of bottle and can detection workflows and robotic separation using an FR5 arm.
- Develop camera-based vision components and deployment solutions for compact platforms including Luckfox, OpenMV and Raspberry Pi, alongside Hikrobot systems.

Student Research Assistant | PLRI, TU Braunschweig

Braunschweig, Germany | November 2022 - September 2025

- Analyzed multi-omics data using network-based methods for NetfLID; supported gene-gene interaction modeling for the Epistasis Disease Atlas.
- Built SVM models and visualizations for biomedical collaborators; supported Linux research infrastructure and contributed to an HPC cluster.

Student Research Assistant | German Aerospace Center DLR

Braunschweig, Germany | November 2024 - June 2025

- Developed CNN-based structural wear models using inspection images and environmental data for aircraft component lifecycle assessment.
- Investigated data provenance and blockchain approaches for lifecycle assessment; coauthored a conference contribution listed by DLR.

Student Assistant | iBMB, TU Braunschweig

Braunschweig, Germany | March 2023 - November 2024

- Prepared experimental data and developed predictive models, including Random Forests, for materials and fire-safety research.
- Built a Django application for internal experimental-data management and researcher access.

Senior Systems Engineer | Infosys

Bengaluru, India | March 2019 - November 2021

- Supported Linux and VMware infrastructure; used SQL for analysis and reporting and Python/Bash scripts for recurring operational tasks.

Selected projects

Industrial visual inspection | Personal research

2026 | Python, PyTorch, CUDA | Case study and recorded examples

- Built a reproducible CLI for data preparation, normal-only calibration, prediction, evaluation and benchmarking; implemented a NumPy baseline and a GPU PatchCore variant.
- Compared preprocessing on a PCB2 development split: cropped 512 px input detected 43/50 defects with 1/50 false alarms, versus 36/50 and 3/50 for full-frame 256 px input.
- Documented weak transfer to a separate component dataset. Results are exploratory development evidence; an independent acceptance evaluation remains pending.

Physics-informed neural networks | Master's thesis

2025 | TensorFlow, Hydra, scientific computing | Repository

- Investigated PINN training for Helmholtz and Allen-Cahn equations using existing frameworks including TensorDiffEq and pinns-tf2.
- Implemented experiments exploring collocation, adaptive weighting and optimization, with saved solution, residual and error visualizations.

Landslide segmentation | Collaborative academic project

2023 | UNet, Swin-based UNet, SegFormer | Team report

- Collaborated with Reiko Lettmoden on multispectral segmentation, comparing architectures, input bands, augmentation and imbalance-aware losses.
- The team report records F1 0.756 for augmented UNet with Lovasz loss versus 0.721 baseline, using a local split of a training subset, not the official benchmark test set.

Additional applied AI and research projects

2022 - 2025 | Portfolio archive

- Speech enhancement: CPC representations and HiFi-GAN in PyTorch; NISQA, PESQ and DNSMOS evaluation against an EffCRN baseline.
- Character AI application: Next.js/TypeScript, Express, MongoDB and OpenRouter integration. Earlier work includes equivariant leakage localization and water forecasting.

Education

M.Sc. Data Science | Technische Universität Braunschweig
October 2021 - June 2025

B.Tech. Electronics and Communication Engineering
Bharati Vidyapeeth's College of Engineering, New Delhi | August 2014 - June 2018

Conference contribution

Janko Kreikemeier, Amit Amit and Steffen Opitz. *Towards a Circular Future with Structural Life 2.0 and Blockchain-Approach.* World Symposium on Materials Sciences and Engineering, Singapore, December 2025. Conference speech; DLR bibliographic record.

Languages

English: C1 | German: A2